Joshua Heinstein

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EDUCATION

Harvey Mudd College — **May 2026**
Bachelor of Science in Engineering — Claremont, CA
Dean's List, GPA: 3.65

Relevant Coursework: Advanced Systems Engineering II, Experimental Engineering, Embedded Systems, Computer Architecture, System-on-Chip Design

EXPERIENCE

Systems Engineer | Infinite Machine - New York, NY — **July 2026**
- Investigated and resolved critical field failures (random vehicle shutoffs) by designing and deploying custom diagnostic hardware to capture field data, identifying the root cause as a software update error.
- Built and launched a fleet of 12 highly reliable data loggers, optimizing the system architecture to guarantee zero data loss during power interruptions and ensure data integrity.
- Developed a custom desktop application with real-time dashboard visualizations, transforming millions of raw data points into readable charts and metrics for engineering analysis.
- Managed the end-to-end product lifecycle of the diagnostic tools, from initial structural design to software development and final field deployment, within a highly compressed 3-week timeline.

Team Lead and Technical Project Manager | XDemics Corporation - Claremont, CA — **Sept 2025 – May 2026**
- Directed a cross-functional team of 6 to automate a complex laboratory process, designing a robotic system to handle sensitive biological materials.
- Analyzed fluid dynamics data and executed rigorous simulation testing to optimize the machine's angles and vibration speeds, achieving a >95% success rate in field testing and saving significant manual labor hours.
- Managed project timelines, task delegation, and technical communication between the student engineering team, faculty advisors, and XDemics' executive leadership (CTO).

Systems Engineer | Amazon Prime Air - Claremont, CA — **Jan 2025 – May 2025**
- Implemented and tested ground and air based hardware/software to reduce drone GPS reliance near tall buildings.
- Designed and executed a data-driven Python algorithm, targeting millimeter-level precision for drone positioning.
- Conducted extensive testing of alternative technologies to mitigate GPS dependency, synthesizing raw performance data into actionable insights.
- Authored a comprehensive 100-page report detailing operational trade-offs, KPIs, and recommendations for senior management.

Structural Engineering Consultant Intern | Kimley Horn - West Palm Beach, FL — **Jun 2024 – Aug 2024**
- Designed over 10 structures for clients, ranging from foundations for industrial machinery, to retaining walls for a marina.
- Inspected the safety of over 30 buildings in Miami-Dade county and submitted detailed reports for each.
- Composed professional calculation packages and drafted CAD blueprints to be signed & sealed by licensed engineers.

Engineering Design and Manufacturing TA, Grader | Harvey Mudd College - Claremont, CA — **Jan 2024 – May 2024**
- Taught foundational teamwork skills to improve the process of creating a final product for a client.
- Provided individual feedback to over 50 students on multiple assignments per week.
- Composed a weekly memo summarizing crucial information that students often overlooked.

SKILLS

Data Analysis & Programming: Python, R, MATLAB, C++, Java, C, Excel, AI/LLM-Assisted Development, Microsoft Suite, Autodesk
Core Competencies: Cross-functional Leadership, Root Cause Analysis, Project Management, Quantitative Research
Languages: English, Spanish (Fluent), German (Conversational)

PROJECTS

Autonomous Environmental Data Collector — **Jan 2024 – May 2024**
- Led a team of 4 to design and deploy an autonomous, GPS-equipped surface robot for remote environmental data collection.
- Utilized MATLAB, Excel, and Python data libraries to process and analyze large sets of collected sensor data, presenting findings and statistical summaries in a final technical report.

Real-Time Audio Algorithm Development — **Nov 2025 – Dec 2025**
- Programmed a highly optimized algorithm (Fast Fourier Transform) to process live audio streams, instantly extracting frequency data.
- Dynamically mapped the quantitative data to a visual display, ensuring ultra-low latency between data intake and visual output.

Microprocessor Quality Assurance & Optimization — **Feb 2025 – May 2025**
- Designed a low-power optimized mathematical computation unit, implementing complex rounding modes.
- Developed custom testing scripts to validate edge cases and ensure rigorous functional accuracy of the unit prior to deployment.